Reproducibility Report for ‘A dual-pathway architecture for stress to disrupt agency and promote habit’ by Giovanniello et al. (2025, Nature)

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2025-12-10

Introduction

[No abstract is needed.] Each reproducibility project will have a straightforward, no frills report of the study and reproducibility results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of your attempt to reproduce the results. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for your reproducibility attempt, and any other essential information. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Giovanniello et al. (2025) investigated the neural mechanisms by which chronic stress disrupts agency learning to potentiate habit formation. In particular, the researchers investigated how distinct nuclei of the amygdala and their projections to the dorsal medial striatum support goal-directed vs habitual behavior, and the disruption of these processes by chronic stress. Chronic, mild stress perturbs these brain regions, rendering a mouse unable to exert volitional control over actions, as measured by two gold-standard assessments of goal-directed behavior: outcome devaluation and contingency degradation.

The focus of this report is to reproduce and simulate data to assess the authors’ core argument that stress disrupts goal-directed action and promotes habits, as indexed by insensitivity to contingency degradation. This behavioral finding is reflected by a Stress x Contingency degradation group interaction on lever press rates.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

The main statistical analysis I will use to reproduce and simulate the main finding is a two-way ANOVA with stress and contingency degradation groups as factors. Lever-press rate during the contingency degradation probe test is what we will use as our measure.

Justification for choice of study

Please describe why you chose to reproduce the results of this study.

The researchers demonstrate the existence of and functional role for CeA-DMS afferents, a circuit I hope to investigate in the context of alcohol-induced aberrant motivational control over behavior. The DMS is a nucleus within the basal ganglia that has a role in the acquisition of goal-directed behavior that supports adaptive decision-making. The CeA is a nucleus within the amygdalar complex that has become a focal point of research on incentive salience. These regions, when investigated on their own, are compromised in disease states such as ethanol dependence, and investigating amygdalar-striatal afferents will provide knowledge on neural mechanisms of disrupted motivational and cognitive control over decision-making.

Anticipated challenges

Do you anticipate running into any challenges when attempting to reproduce these result(s)? If so please, list them here.

A challenge in simulation is capturing the natural variability of animal behavior since performance will follow a normal distribution guided by the means. Therefore, with simulations we lose entropy. The authors use a statistical analysis software called GraphPad Prism. For the purposes of the course and this report, I will use Python packages to reproduce and analyze simulated data. This difference in analysis platforms will yield different values, but should be somewhat similar. There are also many experiments described in this paper. Therefore, I will only focus on one: the effects of chronic stress on a contingency degradation probe test.

Links

Project repository (on Github): https://github.com/tristantd/giovannielloetal2025.git

Original paper (as hosted in your repo): https://github.com/psyc-201/giovannielloetal2025/tree/main/original_paper

Link to Pre-Registration: https://osf.io/registries/drafts/691e0a9f9ef45ebbd997c92b/review

Methods

Description of the steps required to reproduce the results

Please describe all the steps necessary to reproduce the key result(s) of this study.

1. Identify key behavior measure of the experiment to reproduce Assess effects of chronic stress on action-outcome learning using lever press rate during degraded and non-degraded contingencies.
2. Extract descriptive stats for each mouse in each group (press rates during degradation test) I will use available data included with the published paper to reproduce key statistics on Python.
3. Reanalyze using two-way ANOVA, Stress x Contingency degradation group The primary confirmatory analysis will be a Stress x Contingency degradation group interaction. We should also see an effect of contingency degradation and no effect of stress. This corresponds to figure 1i of the original paper.

4. Compare reproduced result to original reported values (F statistic, p-values, effect sizes)

Description of the steps required to simulate and analyze simulated dataset

The hypothesis I want to confirm is that chronic stress will reduce sensitivity to the action-outcome contingency. This would suggest stress mice have disrupted goal-directed control. To evaluate the effects of chronic stress on goal-directed actions, I will simulate behavioral data for 29 mice (7 control, non-degraded; 7 control, degraded; 7 stress, non-degraded; and 8 stress, degraded). These are the exact sample sizes for each group reported in the original paper.

The parameter estimates I will use are the means, SDs, SEMs for each mouse's press rates. Data simulation follows a normal distribution. The design is a 2 (control vs stress) x 2 (degraded vs nondegraded) between subjects ANOVA.

1. Use the original group means and SDs to parameterize the simulated data.
2. Generate simulated behavioral data from a normal distribution for each group.
3. Run the same statistical test as original authors, two-way ANOVA on press rate.
4. Visualization + Interpretation Create a bar graph comparing original and simulated findings side-by-side. Confirm whether the simulated data reproduce insensitivity to contingency degradation in stress mice.

Differences from original study

Explicitly describe known differences in the analysis pipeline between the original paper and yours (e.g., computing environment). The goal, of course, is to minimize those differences, but differences may occur. Also, note whether such differences are anticipated to influence your ability to reproduce the original results.

The differences from the original study are that this will be a simulation by which no behavioral data will be collected. Additionally, this paper uses a statistical analysis software called GraphPad Prism. For the purposes of this course, I will reproduce and simulate, as well as analyze the data using Python. I will use the same analysis, two-way ANOVA, for the simulated data. These differences will potentially influence my ability to reproduce the original results. For example, the different statistical analysis softwares may result in different reported values.

Project Progress Check 1

Measure of success

Please describe the outcome measure for the success or failure of your reproduction and how this outcome will be computed.

The outcome measure for the success of this reproducibility project will be if I can show the same effect that the authors claim in their paper (Stress x Contingency degradation group interaction). This will be comparing my simulated data with the authors' real reported data. This will be computed using an ANOVA on the simulated data and comparing effect sizes.

Pipeline progress

Earlier in this report, you described the steps necessary to reproduce the key result(s) of this study. Please describe your progress on each of these steps (e.g., data preprocessing, model fitting, model evaluation).

Results

Data preparation

Data preparation following the analysis plan.

```
### Data Preparation

#### Load Relevant Libraries and Functions

#### Import data

#### Data exclusion / filtering

#### Prepare data for analysis - create columns etc.
```

```
# Simulation Code Here

# Simulate Figure 1h and 1i from Giovanniello et al. (Nature, 2025)

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pingouin as pg
import statsmodels.formula.api as smf

# Set random seed
np.random.seed(50)

# PARAMETERS / MOUSE GROUPS AND SESSION

n_mice = {
    'control_non_deg': 7, # 7 mice in control, non-degraded group
    'control_deg': 7, # 7 mice in control, degraded group
    'stress_non_deg': 7, # 7 mice in stress, non-degraded group
    'stress_deg': 8 # 8 mice in stress, degraded group
}
sessions = np.arange(1, 5) # 4 training sessions

print(n_mice)
print(sessions)
```

```python
# Group means ± SDs from paper

training_data = { # these are training session means and SDs from the paper's raw data, I got these numbers from uploading
                  # the raw data onto JASP
    1: {'control_deg': (2.140, 0.619), # Day 1 mean and SD for control, degraded group from paper raw data
        'control_non_deg': (1.630, 0.755), # Day 1 mean and SD for control, non-degraded group from paper raw data
        'stress_deg': (2.235, 0.666), # Day 1 mean and SD for stress, degraded group from paper raw data
        'stress_non_deg': (2.626, 1.056)}, # Day 1 mean and SD for stress, non-degraded group from paper raw data
    2: {'control_deg': (5.727, 2.088), # Day 2 mean and SD for control, degraded group from paper raw data
        'control_non_deg': (6.240, 3.677), # Day 2 mean and SD for control, non-degraded group from paper raw data
        'stress_deg': (4.484, 2.296), # Day 2 mean and SD for stress, degraded group from paper raw data
        'stress_non_deg': (4.899, 2.223)}, # Day 2 mean and SD for stress, non-degraded group from paper raw data
    3: {'control_deg': (10.28, 2.460), # Day 3 mean and SD for control, degraded group from paper raw data
        'control_non_deg': (9.104, 2.770), # Day 3 mean and SD for control, non-degraded group from paper raw data
        'stress_deg': (10.20, 2.811), # Day 3 mean and SD for stress, degraded group from paper raw data
        'stress_non_deg': (10.98, 3.392)}, # Day 3 mean and SD for stress, non-degraded group from paper raw data
    4: {'control_deg': (15.37, 2.336), # Day 4 mean and SD for control, degraded group from paper raw data
        'control_non_deg': (16.69, 5.146), # Day 4 mean and SD for control, non-degraded group from paper raw data
        'stress_deg': (14.98, 4.992), # Day 4 mean and SD for stress, degraded group from paper raw data
        'stress_non_deg': (17.38, 6.069)} # Day 4 mean and SD for stress, non-degraded group from paper raw data
}

degradation_data = { # these are degradation session means and SDs from the paper's raw data, I got these numbers from
                     # uploading the raw data onto JASP
    'Control Non-degraded': {'mean': 15.371, 'sd': 5.401, 'n': 7},
    'Control Degraded': {'mean': 3.486, 'sd': 1.852, 'n': 7},
    'Stress Non-degraded': {'mean': 12.225, 'sd': 5.537, 'n': 7},
    'Stress Degraded': {'mean': 11.714, 'sd': 4.730, 'n': 8}
}
```

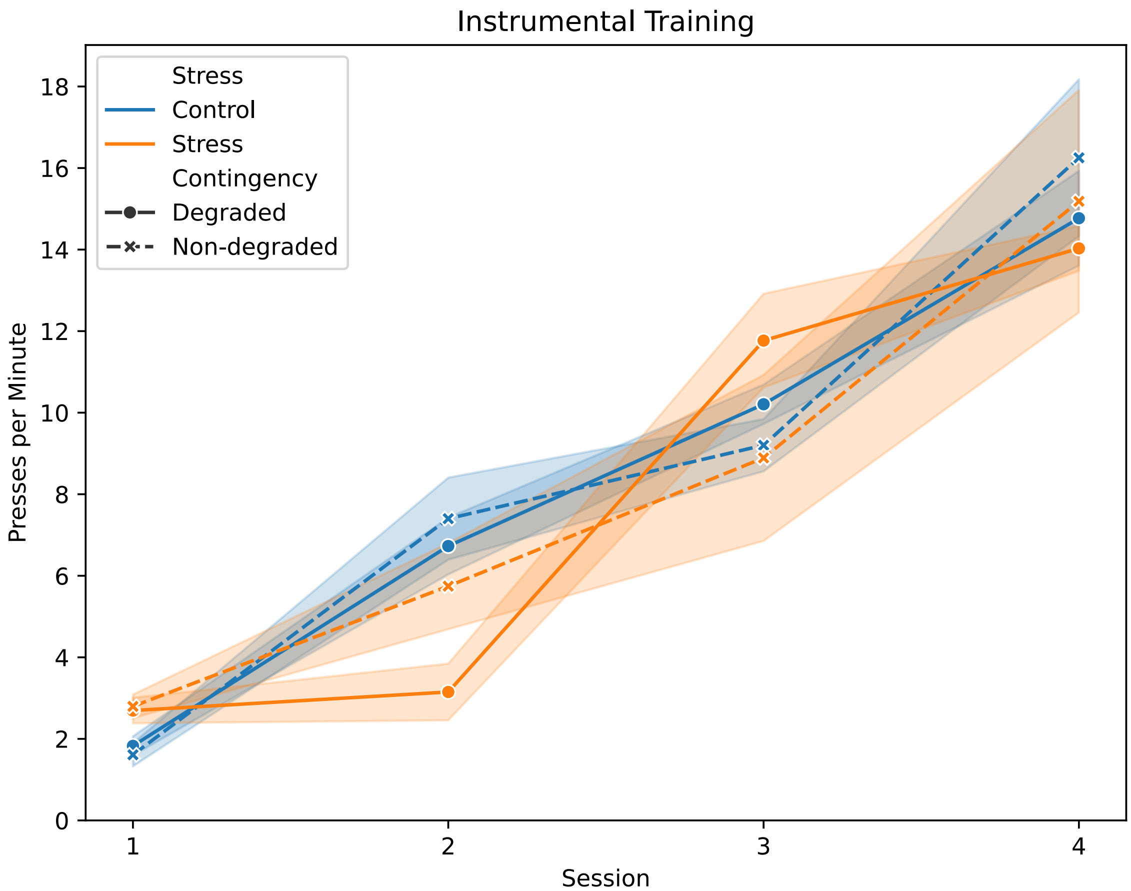
```
# SIMULATE TRAINING DATA (Fig. 1h)
# From the means ± SDs

training_sim = [] # to hold training data from simulation

# Create stable subject IDs
subjects_by_group = {
    group: [f"{group}_{i}" for i in range(n)]
    for group, n in n_mice.items()
}

for session, vals in training_data.items():
    for group, (mean, sd) in vals.items():
        subjects = subjects_by_group[group]
        # Simulate press rates
        rates = np.random.normal(mean, sd, len(subjects))
        # Remove negative values (press rates can't be negative)
        rates = np.clip(rates, 0, None)

        stress = 'Control' if 'control' in group else 'Stress'
        contingency = 'Degraded' if ('deg' in group and 'non' not in group)
else 'Non-degraded'

        for subj, rate in zip(subjects, rates):
            training_sim.append({
                'Session': session,
                'Subject': subj,
                'PressRate': rate,
                'Stress': stress,
                'Contingency': contingency
            })

training_df = pd.DataFrame(training_sim)
print(training_df.head(40))

# Save training data to CSV
training_df.to_csv('output.csv', index=False)

# PLOT SIMULATED TRAINING DATA

plt.figure(figsize=(8, 6))
sns.lineplot(
    data=training_df,
    x='Session', y='PressRate',
    hue='Stress', style='Contingency',
    markers=True, errorbar=('se'),
)
```

```
plt.title("Instrumental Training")
plt.xticks([1, 2, 3, 4])
plt.yticks([0, 2, 4, 6, 8, 10, 12, 14, 16, 18])
plt.ylabel("Presses per Minute")
plt.show()


# REPEATED MEASURES ANOVA for Training Data
# Mixed ANOVA with Session as within-subjects factor and Stress/Contingency as
between-subjects factors

# Create a combined Group factor for mixed_anova (which only supports one
between factor)
training_df['Group'] = training_df['Stress'] + ' ' +
training_df['Contingency']

anova = pg.mixed_anova(
    dv='PressRate',
    within='Session',
    between='Group',
    subject='Subject',
    data=training_df
)

print(anova)

# SIMULATE CONTINGENCY DEGRADATION DATA (Fig. 1i)
# From real means ± SDs

degradation_sim = [] # to hold lever press degradation data from simulation

for group, params in degradation_data.items():
    mean, sd, n = params['mean'], params['sd'], params['n']
    rates = np.random.normal(mean, sd, n) # simulate press rates for
degradation groups
    # Remove negative values
    rates = np.clip(rates, 0, None)

    stress = 'Control' if 'Control' in group else 'Stress'
    contingency = 'Degraded' if 'Degraded' in group else 'Non-degraded'

    for i, r in enumerate(rates):
        degradation_sim.append({
            'Subject': f"{group.replace(' ', '_')}_{i}",
            'Group': group,
            'Stress': stress,
            'Contingency': contingency,
```

```python
            'PressRate': r
        })

degradation_df = pd.DataFrame(degradation_sim) # create dataframe
print(degradation_df.head(20)) # sanity check

# Save degradation data to CSV for analysis on JASP
degradation_df.to_csv('degradation_output.csv', index=False)

# PLOT SIMULATED CONTINGENCY DEGRADATION DATA

plt.figure(figsize=(10, 10))
sns.barplot(
    data=degradation_df, x='Group', y='PressRate',
    capsize=.1, errorbar='se', color='black'
)

sns.stripplot(
    data=degradation_df, x='Group', y='PressRate',
    jitter=True, alpha=0.6, color='grey', size=5
)
plt.title("Contingency Degradation Test")
plt.ylabel("Presses per Minute")
plt.xticks(rotation=40)
plt.tight_layout()
plt.show()


# TWO-WAY ANOVA for Degradation Data
# Between-subjects factors: Stress and Contingency

# Contingency degradation two-way ANOVA
# Using Stress and Contingency as separate factors (not Group, which is
redundant)

deg_anova = pg.anova(
    data=degradation_df,
    dv='PressRate',
    between=['Stress', 'Contingency']
)

print(deg_anova)
```

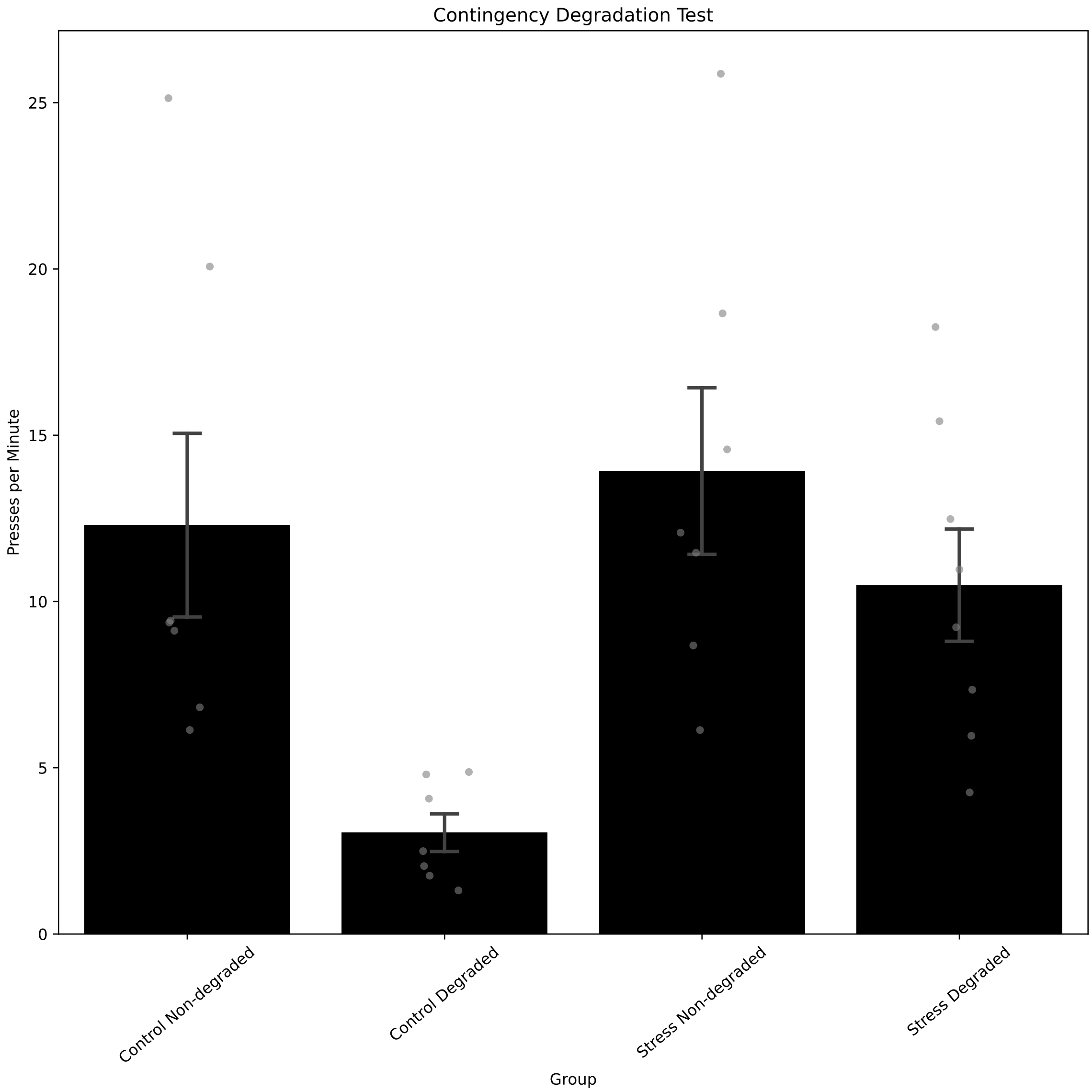
```
{'control_non_deg': 7, 'control_deg': 7, 'stress_non_deg': 7, 'stress_deg': 8}
[1 2 3 4]
    Session            Subject  PressRate   Stress   Contingency
0         1      control_deg_0   1.174142  Control      Degraded
```

```
1         1      control_deg_1   2.120825  Control      Degraded
2         1      control_deg_2   1.755645  Control      Degraded
3         1      control_deg_3   1.233425  Control      Degraded
4         1      control_deg_4   3.013995  Control      Degraded
5         1      control_deg_5   1.844903  Control      Degraded
6         1      control_deg_6   1.656890  Control      Degraded
7         1  control_non_deg_0   2.438052  Control  Non-degraded
8         1  control_non_deg_1   0.661869  Control  Non-degraded
9         1  control_non_deg_2   0.627753  Control  Non-degraded
10        1  control_non_deg_3   1.725385  Control  Non-degraded
11        1  control_non_deg_4   2.280956  Control  Non-degraded
12        1  control_non_deg_5   2.156036  Control  Non-degraded
13        1  control_non_deg_6   1.377403  Control  Non-degraded
14        1       stress_deg_0   1.570648   Stress      Degraded
15        1       stress_deg_1   3.299873   Stress      Degraded
16        1       stress_deg_2   4.442174   Stress      Degraded
17        1       stress_deg_3   2.892855   Stress      Degraded
18        1       stress_deg_4   2.317495   Stress      Degraded
19        1       stress_deg_5   2.729695   Stress      Degraded
20        1       stress_deg_6   1.972625   Stress      Degraded
21        1       stress_deg_7   2.333645   Stress      Degraded
22        1   stress_non_deg_0   2.190680   Stress  Non-degraded
23        1   stress_non_deg_1   2.456285   Stress  Non-degraded
24        1   stress_non_deg_2   2.773345   Stress  Non-degraded
25        1   stress_non_deg_3   2.927456   Stress  Non-degraded
26        1   stress_non_deg_4   2.328987   Stress  Non-degraded
27        1   stress_non_deg_5   4.432718   Stress  Non-degraded
28        1   stress_non_deg_6   2.467846   Stress  Non-degraded
29        2      control_deg_0   7.168360  Control      Degraded
30        2      control_deg_1   8.013797  Control      Degraded
31        2      control_deg_2   8.521597  Control      Degraded
32        2      control_deg_3   2.868566  Control      Degraded
33        2      control_deg_4   6.742661  Control      Degraded
34        2      control_deg_5   7.300353  Control      Degraded
35        2      control_deg_6   6.485914  Control      Degraded
36        2  control_non_deg_0   5.082810  Control  Non-degraded
37        2  control_non_deg_1  11.289555  Control  Non-degraded
38        2  control_non_deg_2   3.944018  Control  Non-degraded
39        2  control_non_deg_3   7.621647  Control  Non-degraded
```




```
        Source           SS  DF1  DF2          MS          F         p-unc  \
0        Group     8.188973    3   25    2.729658   0.347850  7.909933e-01
1      Session  2665.349360    3   75  888.449787  86.297506  2.949194e-24
2  Interaction   136.548640    9   75   15.172071   1.473704  1.733743e-01

      p-GG-corr       np2      eps sphericity   W-spher   p-spher
0           NaN  0.040069      NaN        NaN       NaN       NaN
1  1.011927e-18  0.775377  0.72606      False  0.384348  0.000111
2           NaN  0.150270      NaN        NaN       NaN       NaN
                 Subject                Group   Stress   Contingency  \
0   Control_Non-degraded_0  Control Non-degraded  Control  Non-degraded
1   Control_Non-degraded_1  Control Non-degraded  Control  Non-degraded
2   Control_Non-degraded_2  Control Non-degraded  Control  Non-degraded
3   Control_Non-degraded_3  Control Non-degraded  Control  Non-degraded
4   Control_Non-degraded_4  Control Non-degraded  Control  Non-degraded
5   Control_Non-degraded_5  Control Non-degraded  Control  Non-degraded
6   Control_Non-degraded_6  Control Non-degraded  Control  Non-degraded
7       Control_Degraded_0      Control Degraded  Control      Degraded
8       Control_Degraded_1      Control Degraded  Control      Degraded
```

```
9        Control_Degraded_2      Control Degraded  Control      Degraded
10       Control_Degraded_3      Control Degraded  Control      Degraded
11       Control_Degraded_4      Control Degraded  Control      Degraded
12       Control_Degraded_5      Control Degraded  Control      Degraded
13       Control_Degraded_6      Control Degraded  Control      Degraded
14   Stress_Non-degraded_0   Stress Non-degraded   Stress  Non-degraded
15   Stress_Non-degraded_1   Stress Non-degraded   Stress  Non-degraded
16   Stress_Non-degraded_2   Stress Non-degraded   Stress  Non-degraded
17   Stress_Non-degraded_3   Stress Non-degraded   Stress  Non-degraded
18   Stress_Non-degraded_4   Stress Non-degraded   Stress  Non-degraded
19   Stress_Non-degraded_5   Stress Non-degraded   Stress  Non-degraded

    PressRate
0    6.136125
1    6.817603
2    9.427255
3    9.120652
4   25.136250
5    9.369524
6   20.072113
7    4.873098
8    4.800054
9    4.071509
10   2.044747
11   2.493399
12   1.312063
13   1.754757
14   6.134080
15  12.069998
16   8.677876
17  14.573965
18  18.663771
19  25.870229
```




```
                Source          SS    DF          MS         F     p-unc  \
0               Stress  154.795871   1.0  154.795871  5.115640  0.032660
1          Contingency  282.246118   1.0  282.246118  9.327572  0.005300
2  Stress * Contingency   61.052117   1.0   61.052117  2.017629  0.167834
3             Residual  756.483344  25.0   30.259334       NaN       NaN

        np2
0  0.169867
1  0.271722
2  0.074678
3       NaN
```

Key analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Reproduction Attempt

Open the discussion section with a paragraph summarizing the primary result from the key analysis and assess whether you successfully reproduced it, partially reproduced it, or failed to reproduce it.

Giovanniello et al. (2025) report that chronic, mild stress disrupts action-outcome learning and potentiates habit formation, reflected by a Stress x Contingency degradation group interaction $F(1,25) = 12.75$, $p = 0.002$. I was able to reproduce this statistical result from the available data, $F(1,25) = 12.75$, $p < 0.002$, np2 = 0.34. Using simulated data based on the same group statistics and sample sizes, I observed the interaction, $F(1,25) = 34.59$, $p < 0.001$, np2 = 0.81. Although the exact values differed, performance patterns were comparable. The exact values differed due to the normal distribution imposed during the simulation, but the behavioral pattern was consistent with the original. This indicates a successful reproduction of the authors' argument—stress shifts behavior away from goal-directed control toward habit-like responding. Because of this simulation and low entropy, the effect size I observed is quite large.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis of the dataset, (b) assessment of the meaning of the successful or unsuccessful reproducibility attempt - e.g., for a failure to reproduce the original findings, are the differences between original and present analyses ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the reproducibility attempt (if you contacted them). None of these need to be long.

Although the reproducibility attempt was successful, it is important to consider that there are limitations to simulation-based analyses and that these approaches simplify the biological context. Real animal behavior typically shows heterogeneity driven by motivation, stress exposure variability, reinforcement history, and learning rate differences. Simulating data from normal distributions will reduce noise and inflate effect sizes, overestimating group separability.

Additionally, the real experiment was embedded within a chronic stress protocol that may introduce compounding physiological factors not captured in simulated data. Nonetheless, I was able to reproduce the key finding of stress-induced impairment to adaptive decision-making. This supports the original conclusion and provides confidence in using CeA-DMS circuitry as a target for future investigations, especially in my field, whereby alcohol dependence disrupts motivational control. For reproducibility projects, you should also answer the following four questions – these are for the interpretation/conclusion portion, which otherwise doesn't have the same content as the replication project. 1. What assumptions does your generative model make about participant behavior, and which aspects of the real experiment does it simplify or omit? This simulation project assumes normally distributed behavior within groups, consistent instrumental response acquisition within conditions, and uniform effects of stress. This simulation omits motivational processes, physiological consequences of chronic stress (HPA axis regulation), and learning dynamics that shape the behavioral trajectories (i.e., sustained press rate across days). 2. Show how your simulated dataset flows through your planned analysis pipeline. Does this recover any mismatches or bugs? Upon data simulation, I needed to adjust the columns for different factors in accordance with Python packages to analyze that data. Essentially, each factor needed its own column. The original paper did not have to do this. They used GraphPad Prism in which they could copy and paste their values onto a sheet. My pre-processing resolved bugs when attempting to use analysis packages in Python. My analysis pipeline is aligned with the logic of the original paper, though extra steps were necessary. 3. Identify one insight your simulation gave you about the strengths or limitations of the original experimental design. This simulation highlighted to me that variability in behavior can control the effect we can observe. Although I used the SDs from the original paper, my simulation was constrained to a normal distribution, causing the interaction effect to be larger. 4. How would this simulation help you design a follow-up experiment with a similar paradigm? The simulation estimates effect magnitude and variance that can help with experimental design (e.g., power calculation and sample size). It suggests that future experiments need to incorporate individual difference measurements to include variability that is absent in the model. Follow-up experimental designs could incorporate session-level models to assess changes in learning strategy across sessions, and this can improve sensitivity to subtle shifts in goal-directed control.